

Neural Word Vector Representations Report

Language Modelling 2025–2026 · Master in Artificial Intelligence

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1. Introduction & objectives

This lab implements neural word embeddings with a Continuous Bag-of-Words (CBOW) model trained on English news corpora (30K and 100K sentences). We evaluate embeddings in two ways:

1. Qualitatively — cosine similarity among target words and t-SNE visualizations before/after training.
2. Quantitatively — Reuters topic classification (46 classes) using a CNN, comparing random, frozen, and fine-tuned embedding initializations, plus a GloVe public baseline.

The goal is to study how corpus size, context window, embedding dimension, and training duration affect both intrinsic word-vector quality and downstream classification accuracy.

2. How to run the code

Step	Action
Environment	Intstall packages from <code>environment.yml</code> into your virtual environment
Full run	Set <code>DEBUG = False</code> in Setup, then run all cells top-to-bottom
Quick test	<code>DEBUG = True</code> — 1K sentences, small vocab, 1 epoch

Corpora (under `eng_news_2024/`):

- `eng_news_2024_30k_sentences.txt`
- `eng_news_2024_100K-sentences.txt`

3. Data & preprocessing

Corpora. We use two news datasets with similar average length (~117 characters, ~20 tokens per sentence). Raw token diversity is ~79K (30K file) and ~172K (100K file) before vocabulary capping.

Corpus	Sentences	Avg. tokens	Unique tokens (raw)
30K	30,000	19.7	79,520
100K	100,000	19.8	172,454

Preprocessing (`clean_sentence`): lowercasing, whitespace normalization, digits → `<num>`, Unicode NFKC. Sentences with fewer than 5 tokens are dropped.

Tokenization. `TextVectorization` with `max_tokens` 30K (30K corpus) or 50K (100K corpus). Vocabulary and OOV checks for `target_words.txt` are saved under `artifacts/tables/`.

CBOW samples. For each target token, context words within window size n (2, 5, or 8) form input; target is the center word. Train/validation split: 90/10.

4. Model & training

CBOW architecture (Keras Functional API).

- `Embedding(vocab × d)` → mean over context indices → `Dense(hidden, relu)` → `Dense(vocab, softmax)`
- Full-mode defaults: `d ∈ {100, 300}`, `hidden = 128`, batch 256, **5 CBOW epochs**, EarlyStopping on `val_loss` (patience 2)

Experiment grid (8 runs, `DEBUG=False`): 3 configs on 30K + 5 on 100K varying window (2/5/8), dimension (100/300), and epochs (5/10). Each run logs a row to `artifacts/tables/experiments.csv` and saves `*_E.npy`, `*_word_index.json`, and loss curves.

Best CBOW word-prediction run (validation top-1 accuracy):

`cbow_100k_win2_emb300_ep5_vocab50000` — val top-1 **18.7%**, top-5 **33.9%**.

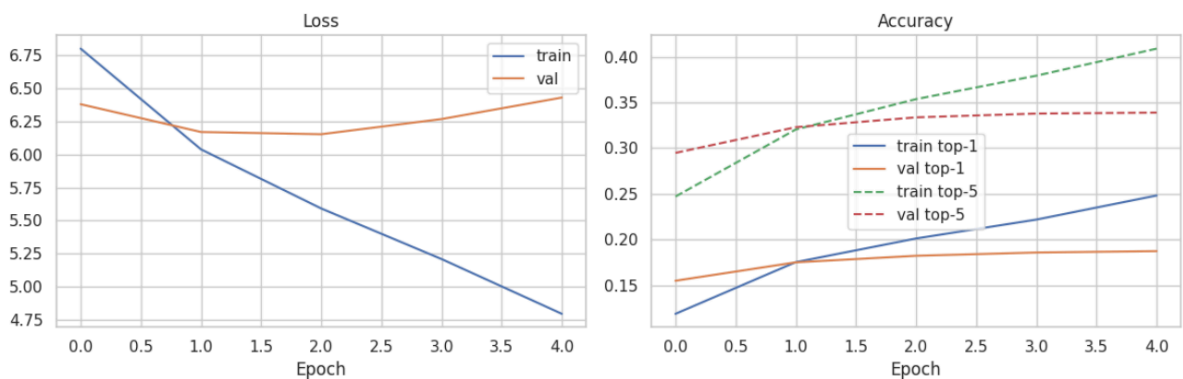


Figure 1. CBOW training/validation loss (representative best run).

5. Qualitative results

Cosine similarity. For each CBOW run we save top-10 neighbors per target word (`artifacts/tables/cosine_<run_id>.csv`). On the best run (`100k`, $d=300$), **calendar words** cluster clearly (*friday* ↔ *monday, thursday*; *september* ↔ *april, march*, ...). **Political terms** group with roles (*president* ↔ *minister, governor*). Some words (*water, music*) have weaker neighbors, consistent with low overall CBOW accuracy.

t-SNE. Target-word plots before vs after training, plus top-500 frequent words with targets highlighted (`*_tsne_before.png`, `*_tsne_after.png`, `*_tsne_top500.png`).

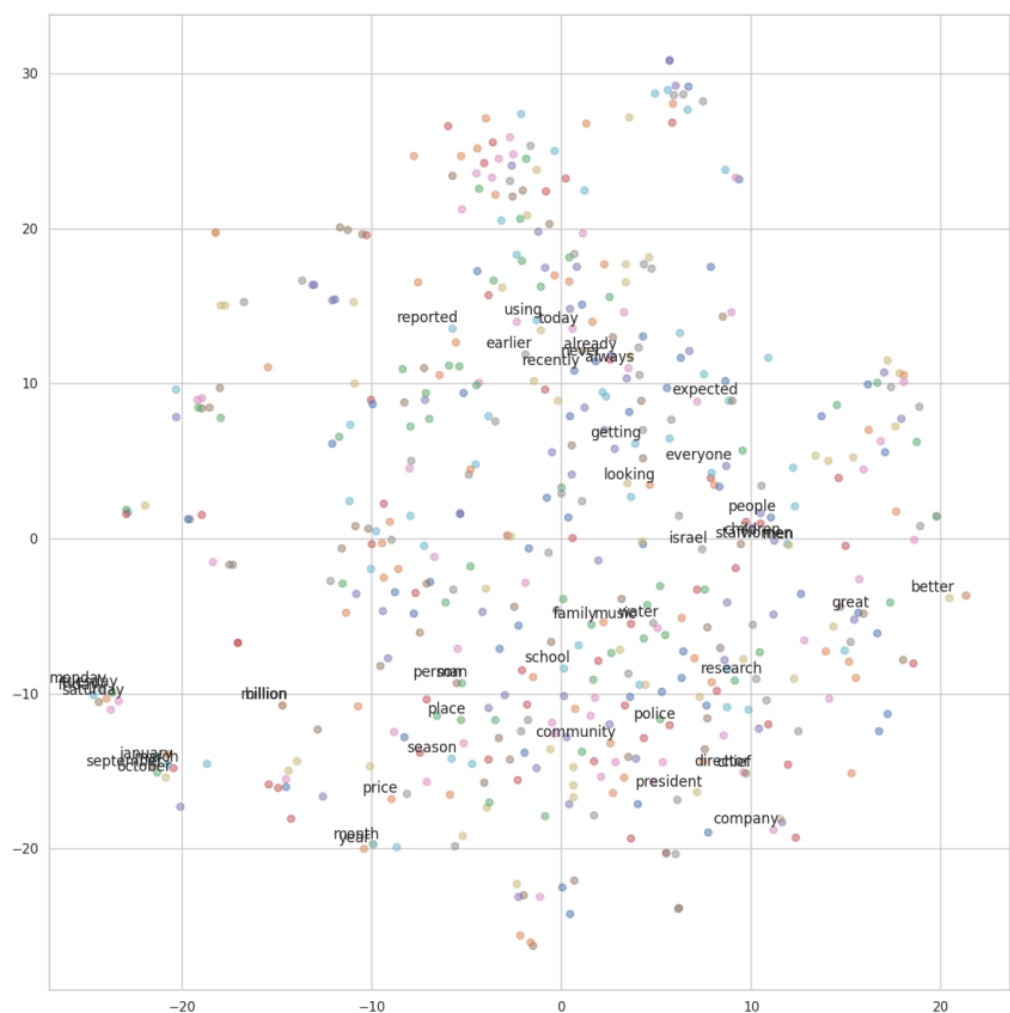


Figure 3. t-SNE of the 500 most frequent vocabulary items (target words annotated).

After training, day-of-week and month names form compact clusters; before training they are scattered. The aggregate top-500 plot shows broader semantic structure while still separating temporal and entity-like groups.

6. Quantitative results

Reuters setup. Keras `reuters` dataset (20K word indices, 46 classes), sequences padded/truncated to 256, CNN classifier (two Conv1D blocks + global max pool + dense). Training: up to 10 epochs, EarlyStopping on `val_loss` / `val_accuracy` (patience 2), batch 128. Three setups per embedding source: baseline (random init), frozen, fine-tune.

Coverage. Fraction of Reuters token IDs ($IDs \geq 3$) that receive a pretrained vector after preprocessing match:

Source	Coverage (approx.)
CBOW 30K vocab	58.9%
CBOW 100K vocab	69.7%
GloVe 100d	89.0%

Reuters test accuracy (selected rows).

run_id	setup	accuracy	macro-F1
baseline_random	baseline	67.5%	0.11
cbow_100k_win2_emb300_ep5	frozen	67.4%	0.11

run_id	setup	accuracy	macro-F1
cbow_100k_win2_emb300_ep5	finetune	75.0%	0.22
cbow_30k_win5_emb100_ep5	finetune	74.9%	0.23
public_glove-wiki-gigaword-100	frozen	72.0%	0.16
public_glove-wiki-gigaword-100	finetune	75.3%	0.19

Full table: [artifacts/tables/reuters_results.csv](#) .

Interpretation.

- Fine-tuning improves accuracy by ~7–12 pp over frozen weights; frozen CBOW alone is near the random baseline because many Reuters IDs stay randomly initialized.
- 100K corpus beats 30K at matched (win=2, d=100) settings: 71.5% vs 69.6% fine-tune.
- Embedding dim 300 beats 100 on 100K (win=2): 75.0% vs 71.5%.
- Window n=2 works best; n=8 is weakest on 100K (73.0% fine-tune).
- Extra CBOW epochs (5 → 10) on 100K win=5 changes Reuters accuracy only marginally (~70.4% both).

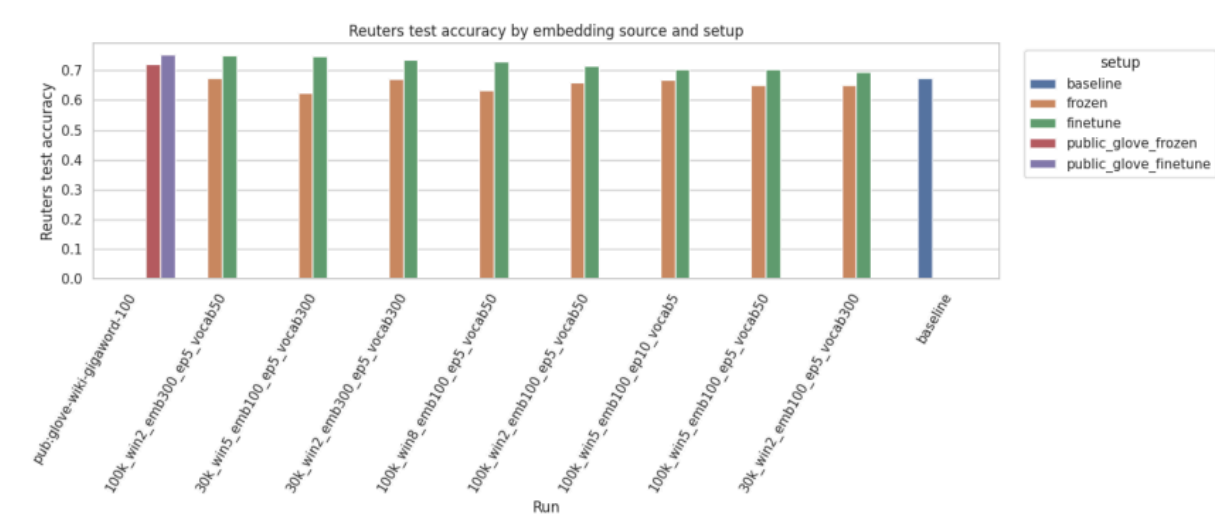


Figure 4. Reuters test accuracy across baseline, frozen, and fine-tuned setups.

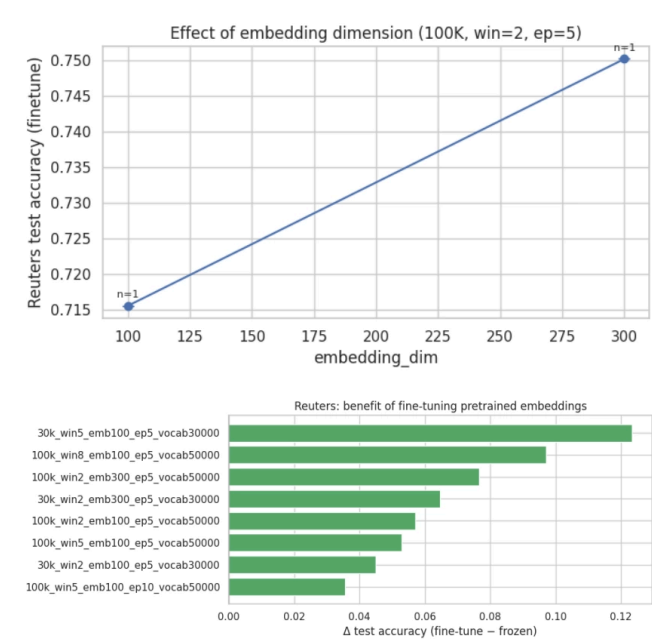


Figure 5. Effect of CBOW embedding size and fine-tuning gain per run

Macro-F1 remains **0.10–0.23** (much lower than accuracy) due to **class imbalance** on Reuters.

7. Public-embedding comparison

We load `glove-wiki-gigaword-100` (gensim) and map vectors into the Reuters embedding matrix (89% coverage). After fine-tuning, GloVe reaches 75.3% test accuracy — on par with our best CBOW (75.0%). Frozen GloVe (72.0%) already beats frozen CBOW, showing the value of large-scale pre-training and better vocabulary overlap with Reuters.

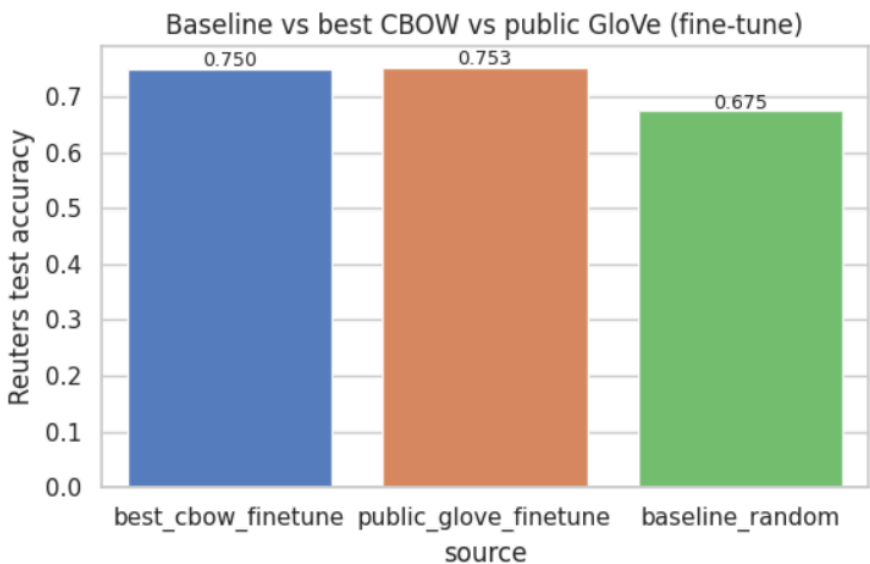


Figure 6. Comparison of random baseline, best CBOW fine-tune, and GloVe fine-tune.

8. Conclusions & limitations

Conclusions

- CBOW on in-domain news produces useful embeddings for Reuters only after CNN fine-tuning; frozen transfer alone is insufficient.
- More data (100K), moderate context ($n=2$), and higher dimension (300) improve both intrinsic and downstream metrics.
- GloVe is a strong baseline mainly because of higher Reuters coverage and pre-training scale; our best CBOW is competitive when fine-tuned.
- Qualitative analysis confirms semantic structure (days, months, political roles) even though CBOW word-level accuracy stays modest (~16–19% val top-1).

Limitations.

- Single downstream task and classifier; results may not generalize to other NLP tasks.
- Imperfect alignment between news-token preprocessing, GloVe, and Reuters token IDs.
- Accuracy overstates performance on rare topics; macro-F1 is low.